Guanxiong Luo, PhD in Computer Science, Machine Learning Research Engineer

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Summary

Machine learning researcher with a PhD in Computer Science, specializing in image processing, computer vision, and computational imaging. Extensive experience developing novel algorithms for image restoration and enhancement, including zero-shot blind deblurring, self-diffusion for inverse problems, and autoregressive image sequence modeling, with publications at CVPR and NeurIPS. Strong foundation in Bayesian inference, optimization methods, and statistical signal processing. Proven ability to implement and optimize algorithms through GPU acceleration and to deliver practical, production-ready solutions through cross-functional collaboration.

Expertise

Computational imaging | **Image processing** | **Computer vision** | **Deep learning Bayesian inference** | **Inverse problems** | **Diffusion models** | **Optimization algorithms** | **GPU acceleration (CUDA)** | **Applied mathematics**

Experiences

Employment

1/2026-present: Independent Researcher — Quantitative Trading & ML

1. Independent research in quantitative trading strategies using probabilistic modeling, time-series analysis, and deep learning.
2. Accepted senior postdoc at Harvard; awaiting US visa. Available for immediate start.

07/2025-12/2025: Postdoctoral Research Fellow at Luxembourg Institute of Health, Luxembourg

1. Develop advanced autoregressive diffusion models for image sequence processing and multi-frame analysis. Design and implement algorithms for temporal prediction, video reconstruction, and 3D volume processing.
2. Collaborate with cross-functional teams to translate research algorithms into practical applications. Focus on productization of machine learning solutions for real-world use cases with optimized GPU implementations.

01/2020-10/2024: Research Scientist at University Medical Center Göttingen, Germany

1. Developed and implemented machine learning models and algorithms for large-scale image data analysis, with focus on computational imaging and computer vision applications.
2. Designed novel algorithms for image reconstruction and restoration, using deep learning, Bayesian methods, and diffusion models.
3. Collaborated with cross-disciplinary teams including software engineers and computational scientists to develop and deploy image processing solutions.
4. Published multiple papers in top-tier venues (NeurIPS, CVPR) and mentored students in research projects.

09/2017-11/2019: Research Assistant at LKS Faculty of Medicine, University of Hong Kong

1. Applied numerical optimization techniques and physics-informed modeling to improve image reconstruction algorithms.
2. Developed Bayesian estimation methods for image reconstruction using generative models as priors, laying foundation for uncertainty-aware image processing.

Education

- **10/2020-11/2023**: PhD in Computer Science, University of Göttingen;
  Advisor: Prof. Dr. Martin Uecker; Thesis: *Development of Advanced Generative Priors for MRI Reconstruction*
- **09/2017-10/2019**: M.Phil in Biomedical Engineering, University of Hong Kong;
  Advisor: Prof. Dr. Cao Peng; Thesis: *The Application of Generative Networks in MR Image Reconstruction*
- **09/2013-07/2017**: B. Eng in Biomedical Engineering, Xi'an Jiaotong University;
  Advisor: Prof. Dr. Junbo Duan; Final Project: *Fast MRI Using Compressed Sensing*.

Award & Honor

- **12/2023**: PhD Graduated with Magna cum Laude, University of Göttingen
- **2017-2019**: Postgraduate Scholarship awarded by The University of Hong Kong
- **06/2017**: Outstanding Graduate of Class 2017 awarded by Xi'an Jiaotong University
- **10/2015**: National Encouragement Scholarship awarded by Xi'an Jiaotong University
- **04/2015**: Meritorious Winner in American Mathematical Contest in Modeling (MCM)

Academic Activities

Talks

- **01/2025**: About *How to use prior information in MR imaging?* at Prof. Dr. Reinhard Heckel's group, Technical University of Munich
- **09/2023**: About *Bayesian MRI reconstruction with joint uncertainty estimation using diffusion priors* at 11th Applied Inverse Problems Conference, Göttingen
- **01/2023**: About *Estimate the uncertainty for MRI reconstruction with learned Bayesian models* at Institute for Numerical and Applied Mathematics, University of Göttingen
- **07/2022**: About *Data Driven Methods for Fast MRI reconstruction* at Cardiac MRI Lab, SJTU, Shanghai
- **09/2021**: About *Bayesian Image Reconstruction with Learned Prior* at Workshop on MRI Acq. & Rec., MGH Harvard

- **05/2021**: About *Using image priors with BART* at ISMRM 2021 Software Session on BART

Teaching and Tutoring

- **WS 2021**: Tutorials for undergraduates and graduates, teaching assistant for a course on deep learning
- **WS 2021**: Teaching assistant for a course on the application of data science to smart city
- **WS 2022**: Tutored one master thesis on MRI reconstruction using deep learning
- **WS 2023**: Tutored one bachelor thesis on MRI reconstruction using diffusion models

Service to the Profession

- Reviews for ICML, NeurIPS, ICLR, CVPR, TMLR, AISTATS, IEEE TMI, IEEE TCI, Medical Physics, Magnetic Resonance in Medicine

Technical Skills & Open-Source Contributions

- **Programming:** Python, C/C++, CUDA, Shell scripting, Linux/Unix | **Deep Learning:** PyTorch, JAX, TensorFlow, TensorBoard
- **Image Processing:** Computational imaging, inverse problems, image restoration, multi-frame analysis | **Dev Toolchain:** Makefile, CMake, Git, GDB, Valgrind, Nsight
- **GPU & Systems:** CUDA kernel optimization, TensorRT inference, multi-GPU training (SLURM, MPI, Docker), HPC cluster computing
- **Selected Repositories:**
  - self-diffusion (PyTorch) — zero-shot image restoration framework
  - aid (PyTorch) — autoregressive diffusion for image sequences
  - spreco (TensorFlow) — diffusion models for image reconstruction
  - mitax (JAX/Flax) — imaging toolbox with GPU acceleration
  - Minimal_Flash_Attention (CUDA) — GPU kernel optimization
  - TensorRT-Cpp-Example (C++/CUDA) — TensorRT inference deployment

Selected Projects

Blind Image Restoration with Self-Diffusion (DeblurSDI)

- **Role:** Lead Researcher and Developer | **Duration:** 2025/06–2026/01
- **Summary:** Developed DeblurSDI, a zero-shot framework for blind image restoration that jointly estimates sharp images and unknown blur kernels without requiring training data or pre-trained models. Uses two coupled neural networks within a self-diffusion process to iteratively refine both image and kernel estimates. Handles diverse degradation types including motion blur and optical aberrations (modeled via Zernike polynomials for realistic lens defects). Achieves state-of-the-art performance across standard benchmarks (33.90 dB PSNR on FFHQ motion deblur vs 25.80 dB for next best method), with demonstrated robustness to kernel size and blur type variation — critical for practical camera image processing. Published at CVPR 2026 (equal contribution).

Zero-Shot Image Restoration (Self-Diffusion)

- **Role:** Lead Researcher and Developer | **Duration:** 2025/02–2025/06
- **Summary:** Developed self-diffusion, a training-free framework for solving general image restoration problems without relying on pre-trained models or external datasets. The method alternates between structured noise injection and self-supervised denoising of an untrained network, exploiting noise-modulated spectral bias to reconstruct images from coarse to fine. Demonstrated across 7 image processing tasks including denoising, deblurring, super-resolution, and inpainting. Achieved competitive or superior results compared to methods using pre-trained models (e.g., 30.30 dB PSNR for $2\times$ super-resolution vs 29.06 dB for baseline). This self-diffusion principle provides the theoretical foundation for the blind imaging extension DeblurSDI (Project 1). Published at NeurIPS 2025.

Autoregressive Image Sequence Generation (AID)

- **Role:** Lead Researcher and Developer | **Duration:** 2024/01–2024/06
- **Summary:** Developed Autoregressive Image Diffusion (AID), a framework combining autoregressive modeling with denoising diffusion for generating temporally coherent image sequences. The model maintains consistency across frames using causal attention mechanisms, significantly reducing hallucinations compared to standard frame-independent diffusion while demonstrating cold-start robustness — generating coherent sequences from minimal initialization. Applicable to video frame interpolation, temporal super-resolution, and 3D image reconstruction. Supports both pixel-space and latent-space (VQVAE-compressed) operation for varying sequence lengths. Published at NeurIPS 2024.

Production-Ready Image Processing Tools

- **Role:** Lead Researcher, Developer and Collaborator | **Duration:** 01/2020–03/2024
- **Summary:** Developed the spreco framework (TensorFlow) for training and deploying diffusion models for image reconstruction, featuring MCMC posterior sampling, interruptible checkpointing, and multi-GPU training on HPC clusters. Built the mitax toolbox (JAX/Flax) for GPU-accelerated imaging operators with mixed-precision training and async data pipelines. Managed full workflow including data preprocessing, model training, testing, optimization, and deployment. Implemented GPU acceleration using TensorRT for real-time inference and created CLI tools for seamless integration into production pipelines.
- **Cross-functional Collaboration:** Worked closely with cross-functional teams including software engineers and domain experts to integrate models into production systems. Documented development processes, performance metrics, and deployment procedures, showcasing ability to bridge research and product development.

Bayesian Image Processing with Uncertainty Quantification

- **Role:** Lead Researcher and Developer | **Duration:** 2021/01–2022/12
- **Summary:** Developed a Bayesian inference framework for image processing with uncertainty estimation using diffusion models. The approach provides trustworthy reconstructions with quantified confidence levels, applicable to arbitrary imaging forward operators and noisy observations. Uses Markov Chain Monte Carlo sampling to estimate pixel-wise uncertainty maps alongside optimal reconstructions. Extensive evaluations focused on uncertainty interpretation, noise modeling, and efficient sampling techniques. Published in Magnetic Resonance in Medicine (top-ten cited article in 2023).

Publications

Conference Papers & Preprints

[1] Yanlong Yang, **Guanxiong Luo***. *Self-diffusion Driven Blind Imaging*. Conference on Computer Vision and Pattern Recognition. (Accepted by CVPR 2026, Equal contribution, arxiv:2510.27439)

[2] **Guanxiong Luo**, Shoujin Huang, Yanlong Yang. *Self-diffusion for solving inverse problems*. Advances in Neural Information Processing Systems. Vol 39. Curran Associates, Inc.; 2025. (Accepted by NeurIPS 2025, arxiv:2510.21417)

[3] **Guanxiong Luo**, Shoujin Huang, Martin Uecker. *Autoregressive Image Diffusion: Generation of Image Sequence and Application in MRI*. Advances in Neural Information Processing Systems. Vol 38. Curran Associates, Inc.; 2024. (Accepted by NeurIPS 2024)

[4] Shoujin Huang, **Guanxiong Luo***, Xi Wang, Ziran Chen, Yuwan Wang, Huaishui Yang, Pheng-Ann Heng, Lingyan Zhang, Mengye Lyu. *Noise Level Adaptive Diffusion Model for Robust Reconstruction of Accelerated MRI*. MICCAI 2024: 498-508. (Equal contribution)

[5] **Guanxiong Luo**, Mengmeng Kuang, Peng Cao. *Generalized Deep Learning-based Proximal Gradient Descent for MR Reconstruction*. International Conference on Artificial Intelligence in Medicine. Springer Nature Switzerland; 2023: 239-244.

Journal Articles

[6] Shoujin Huang, **Guanxiong Luo**, Yuwan Wang, Kexin Yang, Lingyan Zhang, Jingzhe Liu, Hua Guo, Min Wang, Mengye Lyu. *Robust Simultaneous Multislice MRI Reconstruction Using Deep Generative Priors*, Med Image Anal. 2026; 108:103851.

[7] **Guanxiong Luo**, Xiaoqing Wang, Mortiz Blumenthal, Martin Schilling, Erik Hans Ulrich Rauf, Raviteja Kotikalapudi, Niels Focke, Martin Uecker *Generative Priors for MRI Reconstruction Trained from Magnitude-Only Images Uisng Phase Augmentation*, Phil. Trans. R. Soc. A. 2025; 383 (2299): 20240323. (In theme issue on *Generative modelling meets Bayesian inference: a new paradigm for inverse problems*)

[8] Zuojun Wang, **Guanxiong Luo**, Ye Li, Peng Cao. *Using a deep learning prior for accelerating hyperpolarized 13C MRSI on synthetic cancer datasets*. Magn Reson Med. 2024; 92(3): 945-955.

[9] **Guanxiong Luo**, Moritz Blumenthal, Martin Heide, Martin Uecker. *Bayesian MRI Reconstruction with Joint Uncertainty Estimation Using Diffusion Priors*. Magn Reson Med. 2023; 90: 295-311. (In collection of Machine Learning and AI in MRM, top-ten cited articles published in MRM in 2023, featured in the homepage of IBI at TU Graz)

[10] Moritz Blumenthal, **Guanxiong Luo**, Martin Schilling, H. Christian M. Holme, Martin Uecker. *Deep, deep learning with BART*. Magn Reson Med. 2023; 89: 678-693. (Top downloaded MRM article of 2022, featured in the homepage of IBI at TU Graz)

[11] **Guanxiong Luo**, Na Zhao, Wenhao Jiang, Edward S. Hui, Peng Cao. *MRI reconstruction using deep Bayesian estimation*. Magn Reson Med. 2020; 84: 2246-2261.

Conference Abstracts

[12] **Guanxiong Luo**, Moritz Blumenthal, Martin Heide, Martin Uecker *MRI Reconstruction Via Data-Driven Markov Chains With Joint Uncertainty Estimation*. Proceedings of the Annual Meeting of ISMRM. Vol 31. ISMRM; 2023: 0990. (Oral presentation)

[13] **Guanxiong Luo**, Martin Heide, Martin Uecker. *Using data-driven Markov chains for MRI reconstruction with Joint Uncertainty Estimation*. Proceedings of the Annual Meeting of ISMRM. Vol 30. ISMRM; 2022: 0298. (Power Pitch)

[14] **Guanxiong Luo**, Moritz Blumenthal, Xiaoqing Wang, Martin Uecker. *All you need are DICOM images*. Proceedings of the Annual Meeting of ISMRM. Vol 30. ISMRM; 2022: 1510.

[15] **Guanxiong Luo**, Xiaoqing Wang, Volkert Roeloffs, Zhengguo Tan, Martin Uecker. *Joint estimation of coil sensitivities and image content using a deep image prior*. Proceedings of the Annual Meeting of ISMRM. Virtual Conference. Vol 29. ISMRM; 2021: 0280. (Oral presentation)

[16] **Guanxiong Luo**, Moritz Blumenthal, Martin Uecker. *Using data-driven image priors for image reconstruction with BART*. Proceedings of the Annual Meeting of ISMRM. Virtual Conference. Vol 29. ISMRM; 2021: 1756.

[17] **Guanxiong Luo**, Cao Peng. *MRI Reconstruction Using Deep Bayesian Inference*. Proceedings of the Annual Meeting of ISMRM. Virtual Conference. Vol 28. ISMRM; 2020: 0996. (Oral presentation)